POLYLOGARITHMS AND ALGEBRAIC K-THEORY SEMINAR

The goal of this seminar, organized by Andrew Blumberg and Soren Galatius, is to understand the preprint [KRS26a], by Kupers, Rudenko, and Sierra.

This tentative schedule may be updated as we go (for instance, adding references). Current version: November 9, 2026.

Week 1 *Introduction.*

Broad overview of what we'll try to cover. Some history and motivation.

Week 2 *Derived indecomposables 1.* (Optional historical perspective.)

Quillen defined of $D_q(B/A, M)$ for a commutative ring A, a commutative A-algebra B, and a B-module M. Sketch his definition using simplicial resolutions, explain a few basic properties [Qui, §2–4]. Then discuss non-unital commutative k-algebras I, and their unitalization $I^+ = I \oplus k.1$. Discuss exactness properties of the functor $I \mapsto D_*(I^+/k, k)$, which is the analogy in Quillen's setting to the E_k-homology functors we'll discuss in the next lecture. If there is time, explain in what sense this is the derived functor of $I \mapsto I/I^2$.

Then discuss the case when k is a field of characteristic zero, as in [Qui, §11]. By Theorem 11.7 of op.cit., the linear dual $D^*(k/I^+, k)$ is canonically identified with the primitives of a co-commutative Hopf algebra, and hence acquires the structure of a graded Lie algebra.

Week 3 *Derived indecomposables 2.*

Briefly recall what a (non-unital) E_k-algebra is, in spaces and other symmetric monoidal categories. Explain "indecomposables" as left adjoint functor to "trivial algebra", following [GKRW25a, §12–13], [Fra13], and/or [Heu24, §6]. Denoting the (derived) indecomposables of a non-unital E_k-algebra A by $Q_{\mathbb{L}}^{E_k}(A)$, define E_k-homology as $H_*^{E_k}(A) = \widetilde{H}_*(Q_{\mathbb{L}}^{E_k})$. Explain the relationship to k-fold bar constructions.

Discuss special features in the case of an E_∞ algebra in $\mathcal{D}(\mathbb{Q})$, in particular that homology of the 1-fold bar construction has the structure of a commutative Hopf algebra, and that the indecomposables of this Hopf algebra (in the sense of Milnor–Moore) can be identified with E_∞-homology, up to a shift. This explains the Lie coalgebra structure on E_∞-homology.

If time permits, the explicit 1-categorical model B^{Com} from [KRS26a, Definition C.13] could be discussed.

Week 4 *The Quillen spectral sequence and the Rognes spectral sequence.*

Given an E_1 algebra M in spaces, for instance a topological monoid, and an E_1 map $M \to \mathbb{N}$, we can regard M as an E_1 algebra in based spaces and the derived indecomposables form an $\mathbb{N}$-graded based space, whose homology groups are the bigraded homology groups $H_{*,*}^{E_1}(M)$. In this situation there is also the usual bar construction $BM = |[p] \mapsto M^p|$, which is just a plain space. Briefly explain (if this wasn't done in the previous talk) that the E_1-map $M \to \mathbb{N}$ gives rise to a filtration on BM whose associated graded is the suspension of $Q_{\mathbb{L}}^{E_1} M$, so there results a spectral sequence $H_{s,t-1}^{E_1}(M) \Rightarrow H_{s+t}(BM)$ in this situation, see for instance [GKRW25a, §13.8]. Explain or state that with field coefficients, this is a spectral sequence of coalgebras, converging to the "cup coproduct" on homology (linear dual to cup product on cohomology).

Then explain the corresponding statement when M is an E_∞ algebra, with group completion $\Omega^\infty M^{\mathrm{gp}}$ for a spectrum M^{gp}. The spectrum homology $H_*(M^{\mathrm{gp}})$ is the abutment of a spectral sequence whose first page is $H^{E_\infty}_{*,*}(M)$. Over a field of characteristic 0, this spectral sequence can be identified, up to a shift, with the Hopf algebra indecomposables of the spectral sequence for the underlying E_1 algebra. For general Hopf algebra reasons, the indecomposables have a canonical Lie coalgebra structure (this goes back to at least Milnor–Moore's paper on Hopf algebras, the linear dual statement is that the primitives in a cocommutative Hopf algebra inherit a Lie algebra structure).

This could also be a good time to state the formula in [KRS26a, §9.4] for the d^1-differential in the Rognes spectral sequence, which uses the Lie cobracket.

We will be especially interested in the case $M = \coprod B\mathrm{GL}_n(F)$ whose group completion is the K-theory spectrum $K(F)$. Summarize some known results about rational K-theory, including Borel's result when F is the ring of integers in a number field or a localization of such a ring. It would be nice to also say something about Borel and Yang's complete calculation of $H^*(B\mathrm{GL}_n(F);\mathbb{Q})$ in [BY94, section 7], but there might not be time for it in this lecture.

Week 5 *The Goncharov Lie coalgebra.*

The first goal of this lecture is to give the definition of $\mathcal{G}(F) = \oplus_n \mathcal{G}_n(F)$, in terms of E_∞ homology of $M = \coprod_n B\mathrm{GL}_n(F)$. State the vanishing result in [GKRW25b, Corollary 6.12], motivating the definition of $\mathcal{G}_n(F)$ as the lowest interesting value of $H^{E_\infty}_{n,*}(M)$.

The rest of this lecture could be spent on a hands-on exploration of $\mathcal{G}_n(F)$ for $n \in \{1, 2\}$, both of which can be deduced from the definition without a lot of theory. See [GKRW25b, §9.2] regarding $\mathcal{G}_2(F) = H^{E_\infty}_{2,3}(M;\mathbb{Q})$ in terms of the Bloch group $B_2(F)$. (This group $B_2(F)$ is denoted $\mathfrak{p}(F)$ and called the "pre-Bloch" group in [GKRW25b], but we will follow [KRS26a] and call it simply the "Bloch group".) The mathematical content in the case $n = 2$ largely goes back to [Sus90]. The connection between $B_2(F)$ and $\mathcal{G}_2(F)$ is explained in a different way in [KRS26a, §7.2], using a presentation of $\mathcal{G}_2(F)$ as a rational vector space which we will discuss later in the seminar.

Week 6 *Polylogarithms and multiple polylogarithms 1.*

Begin by introducing these as holomorphic functions

$$\mathrm{Li}_{n_1,\dots,n_k} : \mathbb{D}^k \to \mathbb{C},$$

where $\mathbb{D} \subset \mathbb{C}$ is the open unit disk and $\mathbb{D}^k = \mathbb{D} \times \cdots \times \mathbb{D}$. Briefly introduce the "iterated integral" $I(a_0; a_1, \dots, a_k; a_{k+1})$ from e.g. [Gon01, eq.(20)] and [Dup21, eq.(38)], and state how the multivalued functions $\mathrm{Li}_{n_1,\dots,n_k}$ arise as special cases of the iterated integral.

Perhaps the rest of this lecture could be about the special case $k = 1$, the so-called "classical polylogarithms" $\mathrm{Li}_n(z)$. In particular, $\mathrm{Li}_1(z) = -\log(1 - z)$ and $\mathrm{Li}_2(z)$ is called the dilogarithm and was introduced by Euler (in order to numerically evaluate $\zeta(2)$ to high precision, apparently; the talk doesn't need to go into historical perspectives, but see for instance [Var06, Chapter 3] if you're interested).

The function $\mathrm{Li}_n(z)$ can be holomorphically continued to a *multi-valued* function on $\mathbb{C}\backslash\{0, 1\}$: each determination of $\mathrm{Li}_n(z) \in \mathbb{C}$ depends on a choice of homotopy class of path in $\mathbb{C}\backslash\{0, 1\}$ from z to a point in the open interval $(0, 1)$. More generally the iterated integral depends on a certain choice of

path. We will not make much actual use of these multivalued "functions", but there are several ways to replace them by single-valued functions. For instance, the real part of $\mathrm{Li}_1(z)$ is $-\log|1-z|$, which is a single-valued real-analytic function $\mathbb{C}\backslash\{1\}\to\mathbb{R}$.

Explain the Bloch–Wigner function $D_2(z)=\mathrm{Im}\big(\mathrm{Li}_2(z)+\mathrm{Li}_1(z)\log|z|\big)$, a single-valued function $\mathbb{C}\backslash\{0,1\}\to\mathbb{R}$ extracted from Li_2, and state the functional equation it satisfies. Define, or recall from a previous lecture, the Bloch group $B_2(\mathbb{C})$ and explain how D_2 gives rise to a function $B_2(\mathbb{C})\to\mathbb{R}$. Possible reference [Zag07, sections 2–4].

If there is time, briefly state the definition of $\mathcal{L}_n(z)$ from [Zag07, §II.4] or [KRS26b, §7.1.2], generalizing the Bloch–Wigner function to higher polylogarithms.

Week 7 *Multiple polylogarithms 2.*

The multiple polylogarithms from the previous talks were defined as complex numbers $\mathrm{Li}_{n_1,\dots,n_k}(z_1,\dots,z_k)$ for each $z_1,\dots,z_k\in\mathbb{C}\backslash\{0,1\}$, well defined up to some ambiguity. Other incarnations of these entities live in different worlds. The goal of this talk is to describe one of those worlds, the Lie coalgebra $\mathcal{L}^{\mathrm{Hod}}$ of "mixed Hodge structures" and a sub Hopf algebra corresponding to mixed Tate objects.

Start with generalities about Lie coalgebras and commutative Hopf algebras over a field k of characteristic zero, for instance starting from Milnor–Moore (dualized): the main fact that we'll need is that any Lie coalgebra L over k is canonically the indecomposables of a commutative Hopf algebra $H=\mathcal{U}(L)$, its universal enveloping coalgebra. The category $\mathcal{M}=\mathrm{Comod}_H$ of comodules over H which are finite-dimensional over k is an abelian category, and $\otimes_k$ can be upgraded to a symmetric monoidal structure $\otimes_{\mathcal{M}}$ on Comod_H, such that the forgetful functor $r:\mathrm{Comod}_H\to\mathrm{Vect}_k$ is symmetric monoidal. In algebraic geometry language, Comod_H is (by definition) the category of representations of the group scheme $G=\mathrm{Spec}(H)$ over $\mathrm{Spec}(k)$. In practice, there are many situations where one *first* defines the category $\mathcal{M}$ and then uses it to extract the Hopf algebra H, equivalently the Lie coalgebra L, for which $\mathcal{M}=\mathrm{Comod}_H$. The reconstruction of H from $(\mathcal{M},\otimes_{\mathcal{M}},r)$ is often summarized as "by the Tannakian formalism", but is actually very concrete, and importantly, it lets one easily *name elements* of H: the Hopf algebra H is generated as a k-vector space by $(M,v,\phi)\in H$ where $M\in\mathcal{M}$, $v:k\to r(M)$, and $\phi:r(M)\to k$, subject to $(M,v,\phi)\in H$ being linear in both v and ϕ, as well as the equivalence relation $(M,v,\phi\circ r(f))\sim(N,r(f)\circ v,\phi)$ for each morphism $f:M\to N$. See for instance [Gon01, Theorem 3.2] or [Dup24, §8.3].

Then briefly explain the category MHS of "mixed Hodge structures", which is an example. State its definition, in terms of two filtrations, and basic properties. It's somewhat surprising that it's an abelian category! Its role in life is to fix into a diagram

$$\begin{array}{ccc}\mathrm{Var}_F & \xrightarrow{\mathcal{H}^j} & \mathrm{MHS}\\ \downarrow & & \downarrow r\\ \mathrm{Top} & \xrightarrow{H^j(-;k)} & \mathrm{Vect}_k,\end{array}$$

where $F\subset\mathbb{C}$ is a subfield (such as $F=\mathbb{C}$), the left vertical arrow sends X to $X(\mathbb{C})$ in the euclidean topology, and $k\in\{\mathbb{Q},\mathbb{R},\mathbb{C}\}$. The category MHS depends on k and we'll mostly take $k=\mathbb{Q}$. A similar factorization

holds for simplicial varieties, relative cohomology of a pair of varieties, and compactly supported cohomology.

Explain the special "Tate" objects $\mathbb{Q}(n) \in \mathrm{MHS}$, one for each $n \in \mathbb{Z}$, and the full subcategory $\mathrm{MHTS} \subset \mathrm{MHS}$ on those objects which are iterated extensions of Tate objects. Then, MHS is comodules over some Hopf algebra H^{MHS}, and the subcategory MHTS is comodules over a sub Hopf algebra $H^{\mathrm{MHTS}} \subset H^{\mathrm{MHS}}$, which by definition is $H^{\mathrm{MHTS}} = \mathcal{U}(\mathcal{L}^{\mathrm{Hod}})$. We can name explicit elements of this graded Hopf algebra by the general theory: in degree m, they look like (M, v, ϕ), with $M \in \mathrm{MHTS}$, $v : \mathbb{Q}(0) \to \mathrm{Gr}_0^W(M)$, and $\phi : \mathrm{Gr}_{-2m}^W(M) \to \mathbb{Q}(m)$.

Additional references: [Del94] for a description of the category of real mixed Hodge structures, which implies that $\mathcal{L}^{\mathrm{MHTS}}$ is a cofree Lie coalgebra.

Week 8 *Multiple polylogarithms 3*

The goal of this talk is to define elements $\mathrm{Li}^{\mathrm{Hod}}_{n_1,\dots,n_k}(z_1, \dots, z_k) \in \mathcal{L}^{\mathrm{Hod}}$, the grade Lie coalgebra such that the category of graded comodules over $\mathcal{U}(\mathcal{L}^{\mathrm{Hod}})$ is equivalent as a symmetric monoidal category to MHTS. These will allow us to parse the explanation around [KRS26a, Theorem D.b].

In order to apply the theory discussed in the previous lecture, we need a good source of varieties X (or simplicial varieties, or pairs of varieties) such that $H^j(X)$ is provably an extension of Tate objects. Perhaps surprisingly, the fundamental group is a useful source (this insight goes back at least to Deligne's long paper [Del89]).

If π is a finitely generated group, then $\mathbb{Q}[\pi]/I^{m+1}$, the group ring modulo the $(m+1)$st power of the augmentation ideal, is a finite-dimensional vector space. It is an iterated extension of the vector spaces I^k/I^{k+1}, so there exists noncanonically an isomorphism $\mathbb{Q}[\pi]/I^{m+1} \cong \oplus_{k=0}^m I^k/I^{k+1}$ in rational vector spaces.

If $\pi = \pi_1(X, a)$ is the fundamental group of a variety X, then there is a formula for $\mathbb{Q}[\pi]/I^{m+1}$ as $H^m(X^m, D)$, for a certain subvariety $D \subset X^m$, see for instance [Gon01, Theorem 4.1] or the main result of [Loo24]. This formula then gives a canonical lift of $\mathbb{Q}[\pi]/I^{m+1}$ to MHS. In many cases, $I/I^2 = H_1(X; \mathbb{Q})$ is isomorphic to a sum of copies of $\mathbb{Q}(1)$, in which case $\mathbb{Q}[\pi]/I^{m+1} \in \mathrm{MHTS} = \mathrm{Comod}_{\mathcal{U}(\mathcal{L}^{\mathrm{Hod}})}$.

The discussion of fundamental group generalizes to $\pi = \pi_{\leqslant 1}(X)(x, y)$, the set of morphisms in the fundamental groupoid of X; in this case, $\mathbb{Q}\pi/I^{m+1} \in \mathrm{MHTS}$ is denoted $\mathcal{P}_M^m(X; x, y)$ in [Gon01, §4.4]. The points x and y can also be "basepoints at ∞", but in this talk it may be advisable to sweep that under the rug.

After this discussion of fundamental groups, explain the meaning of $I^{\mathrm{Hod}}(a_0; a_1, \dots, a_k; a_{k+1}) \in \mathcal{L}^{\mathrm{Hod}}$ as in [Gon01, Definition 5.5]. These iterated integrals are incarnations of the iterated integrals from two weeks ago, of which the multiple polylogarithms $\mathrm{Li}_{n_1,\dots,n_k}$ are special cases. Hence we also get incarnations $\mathrm{Li}^{\mathrm{Hod}}_{n_1,\dots,n_k}(z_1, \dots, z_k) \in \mathcal{L}^{\mathrm{Hod}}$.

If there is any time left, state the connection between this picture and the functions described in earlier lectures: there is a linear function

$$p_{\mathbb{R}} : \mathcal{L}^{\mathrm{Hod}} \to \mathbb{R}$$

sending $\mathrm{Li}^{\mathrm{Hod}}_{n_1,\dots,n_k}(z_1, \dots, z_k) \in \mathcal{L}^{\mathrm{Hod}}$ to a single-valued version of the multiple polylogarithms (for instance, $\mathrm{Li}_2^{\mathrm{Hod}}(z)$ is sent to the Bloch–Wigner function).

Additional references: [Del94] for a description of the category of real mixed Hodge structures. When F is a number field, there is a completely

parallel discussion with "mixed Hodge structure" replaced by "mixed motive", which is discussed in [Dup24].

Week 9 *Steinberg modules 1.*

This talk should give a hands-on perspective on $\mathrm{St}(V)$, $\mathrm{St}^2(V)$, and $\mathrm{St}^\infty(V)$, with emphasis on how one names elements of these vector spaces.

Define the Tits building $T(V)$, state the Solomon–Tits theorem, define $\mathrm{St}(V) = \widetilde{H}_{n-2}(T(V);\mathbb{Q})$, and define apartment classes $[v_1,\dots,v_n] \in \mathrm{St}(V)$ for any ordered basis $(v_1,\dots,v_n)$ of V, and state the presentation in [KRS26a, §2.3.2]. Mention the product $\mathrm{St}(V)\otimes\mathrm{St}(W)\to\mathrm{St}(V\oplus W)$, under which $[v_1,\dots,v_n]$ is the product of the canonical element $[v_j]\in\mathrm{St}(P_j)$, where $P_j = Fv_j \subset V$.

Then state the two perspectives on $\mathrm{St}^2(V)$: as a functor, it is the bar construction of St; and objectwise, it is just $\mathrm{St}^2(V) = \mathrm{St}(V)\otimes\mathrm{St}(V)$. The comparison of these two definitions is an important result of Miller–Patzt–Wilson, and leads to a comparison map $\mathrm{St}(V)\otimes\mathrm{St}(V)\hookrightarrow(\mathrm{B}^{\mathrm{As}}\mathrm{St})_n(V)$ called the associative symbol in [KRS26a, §2.6.3]. (Remark on notation in [KRS26a]: they use "Bar" for an ∞-categorical notion of bar-constructing and B^{As} for the usual 1-categorical model with tensor powers assembled into a simplicial object.) Define the elements $I[v_1,\dots,v_n]\in\mathrm{St}^2(V)$ and state [GKRW25b, Theorem C], which imply that any one of these elements generate $H_0(\mathrm{GL}(V);\mathrm{St}^2(V))\cong\mathbb{Q}$.

Either concurrently with the discussion of St^2 or separately, state the two (equivalent) perspectives on $\mathrm{St}^\infty(V)$: as a quotient of $\mathrm{St}^2(V)$ by the sum over all non-trivial decompositions $V = V_1\oplus V_2$ of the images of the product map $\mathrm{St}^2(V_1)\otimes\mathrm{St}^2(V_2)\to\mathrm{St}^2(V)$, and as the commutative Koszul dual $B^{\mathrm{Com}}(\mathrm{St})$. Define the Steinberg correlators $C[u_0:\cdots:u_n]\in\mathrm{St}^\infty(V)$ for ordered affine bases $(u_0,\dots,u_n)$ of V. (If possible, it would be nice to do St^2 and St^∞ in parallel, which would perhaps require some prerequisites about associative and commutative Koszul duality. I.e., the 1-categorical models B^{As} and B^{Com}; there's some discussion in the appendices of [KRS26a], and perhaps the book of Loday and Vallette is also useful. On the other hand, an extensive detour into Koszul duality theory could perhaps be a distraction in this seminar.)

Additional references: [CRR26, §3.1-2] for more on $\mathrm{St}(V)$ and $\mathrm{St}^2(V)$, [KS21] for a proof of the Solomon–Tits theorem.

Week 10 *Steinberg modules 2.*

This talk should connect the hands-on algebraic perspective from the previous talk with the derived indecomposables and E_k-homology from previous talks. Sources could include [KRS26a] and [GKRW25a]. A goal could be to state [KRS26a, Theorem 4.3] and explain some ingredients in its proof.

This talk will make use of the flexibility of studying E_k-algebras in different underlying symmetric monoidal categories $\mathcal{C}$, we'll especially interested $\mathcal{C}$ being a category of functors out of finite-dimensional F-vector spaces and their isomorphisms, equipped with the Day convolution monoidal structure (called the "Levi tensor product" and denoted $\boxplus$ in [KRS26a]). Following [KRS26a, §4.2–4.4], explain or state how the "split buildings" $\widetilde{D}^k(V)$ show up, using the iterated-bar-construction model for E_k-indecomposables. Mention the "non-split buildings" $D^k(V)$, essentially due to Rognes. The forgetful map $\widetilde{D}^k(V)\to D^k(V)$ is not an equivalence, but state Nesterenko–Suslin's comparison theorem. The result of Miller–Patzt–Wilson and its generalization in [KRS26a, §4.4] identifies $D^k(V)$ with the bar construction

of $D^{k-1}(V)$ when $k \geqslant 2$, and in particular an equivalence between St^2 and the bar construction of St.

Finally, explain the description of the Lie cobracket on $\bigoplus \mathcal{G}_n(F) = \bigoplus H_1(\mathrm{GL}_n(F); \mathrm{St}^\infty)$, following [KRS26a, section 4.5].

Week 11 *Steinberg modules 3.*

This talk will focus on $\mathrm{St}^\infty(V)$.

Explain the resolution of $\mathrm{St}^\infty(V)$ given by [KRS26a, Proposition 5.12]. Following [KRS26a, §5.2], use this resolution to deduce the presentation of $\mathcal{G}_n(F)$ as a vector space, generated by the elements $\mathrm{Cor}^{\mathcal{G}}(a_0 : \cdots : a_n)$. One form of the presentation is in [KRS26a, Proposition 5.15], which is slightly reformulated in [KRS26a, §7.1].

Mention the realization constructions from [KRS26a, Theorem 7.29]. Imprecisely, this tells us that "relations between multiple polylogarithms" is related to the relations in $\mathcal{G}_n(F)$, and hence to algebraic K-theory of F via the Rognes spectral sequence.

Finally, define the depth filtration on $\mathcal{G}_n(F)$ and explain how depth-one corresponds to (one-variable) polylogarithms.

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